

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2024-2025

Comprehensive Examination
(EC-3 Regular)

Course No. : DSECLZG522
Course Title : Big Data Systems
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours + 20 Mins
Date of Exam : 06-09-2025 (EN)

No. of Pages	= 2
No. of Questions	= 8

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q.1. A system has a Mean Time To Failure (MTTF) of 800 hours and a Mean Time To Repair (MTTR) of 8 hours. Calculate the system's availability, given that the system's downtime is solely due to repair time, and then discuss how the relationship between MTTF and MTTR impacts the system's overall reliability. Additionally, using your calculated availability, propose a maintenance scheduling strategy for a real-world data center operation, considering the potential consequences of system downtime on business operations.

[Marks 5]

Q.2. An enterprise company need to address the following challenges with Big Data Analytics:

- a) Struggling with a Java based ELT tool to manage and extract value from the growing volume of data and need to unify information across federated sources. Each disparate MySQL server records around 2 million updates, a total of 50 million updates daily.
- b) Unable to store and relate raw data collected from system logs, sensors, or click streams with customer and line-of-business data managed in enterprise systems.
- c) Each end-to-end ELT process takes 14 hours. It should be completed within an hour.
- d) Some data warehousing queries on structured data are available only in SQL and there is no expertise available for transforming these SQL queries into Hadoop eco system tools.

Recommend suitable solutions for the above challenges by making use of Hadoop Ecosystem products/tools. Give justifications for your recommendations.

[Marks 5]

Q.3. What are the 2 different types of pluggable schedulers that can be used in Hadoop YARN?

Which one of them guarantees maximum utilization of cluster resources? Fair share scheduler is plugged into YARN resource manager with queues Q0, Q1, Q2, and Q3. All these queues are configured to pre-empt containers from other queues. The weights allocated to each of the queues are given below:

Q0 - weight 0.00

Q1 - weight 0.50

Q2 - weight 0.30

Q3 - weight 0.20

What percentage of total resources will be used by the jobs at run time, in each of the following scenarios?

- (a) Job1 is submitted to Q1 and it is the only job running on the cluster
- (b) While Job1 was running in Q1, Job2 is submitted to Q2
- (c) While Job1 was running in Q1 and Job2 was running in Q2, Job3 is submitted to Q3
- (d) While Job1 was running in Q1, Job2 in Q2 and Job3 in Q3, Job4 is submitted to Q0

[Marks 5]

Q.4. Go through the Spark Scala code given below and answer the Questions (a), (b), (c), (d):

```
1 val lines = new Array[String](2)
2 lines(0) = "one one reject"
3 lines(1) = "own own own"
4 val stringRDD=sc.parallelize(lines,2)
5 println("Number of Partitions: "+stringRDD.getNumPartitions)
6 val wordsRDD = stringRDD.flatMap(x => x.split(" "))
7 val wc = wordsRDD.map(word => (word, 1)).reduceByKey(_ + _)
8. wc.collect()
9. val filt1RDD = wordsRDD.filter(x => x.contains("own"))
10. filt1RDD.collect()
```

Questions:

- (a) What is the output at line No. 5
- (b) What is the output at line No. 8
- (c) What is the output at line No.10
- (d) Modify the code to find out the word count after excluding the word "reject"

[Marks 5]

Q.5. You are developing a bigdata analytics application in Pyspark using SparkSQL queries. The data file is structured with the schema defined in the header and it is stored on HDFS in the form of a CSV file. What are the different methods available for populating the contents of the file in an SQL table in your PySpark program for firing the SparkSQL queries? What are the advantages and disadvantages of these methods?

[Marks 5]

Q.6. What is Multiple linear regression (MLR)? Suppose you are working on an MLR project using Spark MLlib, and your dataset consists of high-dimensional data with a mix of numerical and categorical features. Describe how you would apply the concept of feature preparation to optimize the accuracy of your MLR model, and discuss the trade-offs between different feature preparation strategies, such as feature scaling, encoding categorical variables, and dimensionality reduction techniques, in the context of your project.

[Marks 5]

Q.7. Why windowing needed in Stream computing? Data from a smart device is transmitted over TCP/IP connection to a Spark streaming analytics system. Data from the device arrives at the rate of 1 event per second. The batching interval in the streaming context is set to 4 seconds. The stream analytics system uses a stateful algorithm with an analysis window of 64 seconds. It is required to generate an output from the streaming system every 32 seconds. What type of windowing is recommended by you? Give appropriate configuration parameters for the windowing method suggested by you. What will be the number of mini-batch RDDs in the Dstreams generated in one window duration.

[Marks 5]

Q.8. An eCommerce company is planning to launch a new application on cloud for online sale of items. During normal seasons of the year, this application deployed on a cluster of 4 servers will guarantee acceptable response to the customers. During festival seasons, the sales is expected to double.

- (a) Which feature of cloud will be useful in minimizing the data center costs without compromising the response time?
- (b) What are the recommended features of components to be used in the design of this application?
- (c) Explain how the features of NoSQL databases can be leveraged in the design and deployment of this application?

[Marks 5]
